



Lessons from debiasing data for fair and accurate predictive modeling in education

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ABSTRACT

The past few years have witnessed an explosion of attention given to the bias displayed by Machine Learning (ML) techniques towards different groups of people (e.g., female vs. male). Although ML techniques have been widely adopted in education, it remains largely unexplored that to what extent such ML bias manifests itself in this specific setting and how it can be reduced and eliminated. Given the increasing importance of ML techniques in empowering educators to teach effectively, this study aimed to quantify the characteristics of the original datasets that might be correlated with the subsequent predictive unfairness displayed by ML models. To this end, we empirically investigated two types of data biases (i.e., *distribution bias* and *hardness bias*) towards students of different sexes and *first-language backgrounds* across a total of five frequently-performed predictive tasks in education. Then, to improve ML fairness, we drew inspiration from the well-established research in Class Balancing Techniques (CBTs), where samples are generated/removed to alleviate the predictive disparity between different prediction classes. We proposed two simple but effective strategies to empower class balancing techniques for alleviating data biases and improving prediction fairness. Through extensive analyses and evaluations, we demonstrated that ML models may greatly improve prediction fairness (improvement up to 66%) with only a small sacrifice (less than 1%) in prediction accuracy by balancing the training data with the use of students' demographic information and the overall hardness bias measure. All data and code used in this study are publicly accessible via <https://github.com/lsha49/FairEdu>.

1. Introduction

Machine Learning (ML) techniques have been increasingly applied to tackle a variety of predictive tasks in education over the past decade (Feng, Tang, & Liu, 2019; Guo, Sun, Wang, Gao, & Feng, 2019; Li et al., 2023). These tasks include identifying students who are likely to enroll at a higher education institution and enabling the institution to better allocate its scholarships (Slim, Hush, Ojah, & Babbitt, 2018), classifying different types of posts produced by students in discussion forums (e.g., those requiring urgent responses vs. those not) and enabling educators to provide timely help to address students' confusion (Bakharia, 2016), predicting the likelihood of a student successfully completing a course (Halawa, Greene, & Mitchell, 2014; Yu, Wu, & Liu, 2019), forecasting a student's interest and participation in a STEM (science, technology, engineering, and mathematics) career after college study (Mandalapu & Gong, 2019; Yeung & Yeung, 2019), etc. With the aid of ML techniques, educators can effectively perform their teaching duties and provide more personalized learning experiences to students. For instance, by using a logistic regression model to identify students who are likely to drop out of a course, Burgos et al. (2018)

managed to provide timely help to these students and successfully reduced the dropout rate by 14%.

It is worth noting that the construction and application of an ML model in education are mostly centered around the overall model prediction *accuracy* (e.g., to classify as many educational forum posts correctly as possible). However, only achieving high predictive accuracy on the overall population may not be sufficient in an educational context. Long before the adoption of machine learning to support teaching and learning, educational researchers devoted themselves to ensuring equal learning opportunities for students of different social/demographic backgrounds (Leicester, 1996; Malherbe, 2004), which include students of different races, gender identities, religious beliefs, disability status, etc. In this context, adopting an accurate but unfair model that produces accurate predictions for one group of students but less accurate predictions for other groups of students is clearly undesirable (Kizilcec & Lee, 2020; Mazzoli Smith, Todd, & Laing, 2018), since it may result in inadequate educational support and thereby contradict the notion of equal opportunity for learning.

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To our knowledge, only a few studies have attempted to diagnose the ML models used in education from the perspective of prediction fairness (Kizilcec & Lee, 2020; Loukina, Madnani, & Zechner, 2019; Sha, Rakovic, Whitelock-Wainwright, et al., 2021; Yan, Kao, & Ferrara, 2020). These studies were generally limited in the following two aspects: (i) only focusing on the detection of ML bias and making no attempt to reduce such bias (Fazelpour & Danks, 2021); and (ii) were mostly centered around one or two predictive tasks (e.g., student dropout prediction (Karimi-Haghighi, Castillo, Hernandez-Leo, & Oliver, 2021) or forum post classification (Sha, Rakovic, Whitelock-Wainwright, et al., 2021)) and thus the validity and applicability of the findings from one task remained unknown to another task. This calls for a more comprehensive evaluation of the prevalence of ML bias across different educational predictive tasks and the development of effective methods to combat such bias.

Following our previous studies (Sha, Raković, Das, Gašević, & Chen, 2022; Sha, Rakovic, Whitelock-Wainwright, et al., 2021), which demonstrated that data imbalance was one of the main causes for ML bias in the task of classifying educational forum posts and such bias could be potentially reduced with the aid of class balancing techniques (e.g., SMOTE), we aimed to measure and tackle ML bias at the data level. To enable practical insights and guidelines for other researchers to combat prediction bias, we included a total of five common predictive tasks for analyses and evaluation, which covered major milestones of a student's coursework and could essentially be tackled as binary classification problems. These tasks include classifying different types of discussion forum posts which are generated during a course, characterizing a specific type of rhetoric move in student-written assignments, predicting the likelihood of a student dropping out of a course, estimating a student's final course performance, and forecasting whether a student would pursue a STEM career after a college study. In this study, we investigated ML bias towards students with two types of protective attributes, i.e., *sex* (female vs. male) and *first-language background* (English-as-second-language vs. English-as-first-language). To tackle ML bias at the data level, we adopted two metrics to measure biases hidden behind the data used to train a predictive model, namely (i) *distribution bias*, which refers to the unequal distribution of different student groups (e.g., female vs. male) within each prediction task label (e.g., dropout vs. non-dropout); and (ii) *hardness bias*, which measures to what extent the data instances contained in a dataset are difficult to be correctly labeled. Then, we performed a systematic evaluation on the effectiveness of two representative class balancing techniques (i.e., SMOTE for over-sampling and NearMiss for under-sampling) in alleviating data bias and subsequently improving prediction fairness. Furthermore, we proposed to empower the class balancing technique with two simple strategies, one particularly dealing with the distribution bias and the other dealing with the hardness bias in data. Through extensive experiments, we contributed with the following main findings:

- All five datasets used in this study displayed certain biases, which we should raise awareness of before using them to construct predictive ML models;
- The direct use of class balancing techniques cannot guarantee the reduction of data bias, which, on the contrary, tends to exacerbate the prediction bias of ML models;
- The distribution bias of a dataset can be removed by using both prediction task labels and students' demographic labels to balance the dataset, after which the prediction fairness could be generally improved;
- After removing distribution bias, the prediction fairness could be further boosted by incorporating the hardness bias measure to perform sample selection with a minor sacrifice (less than 1%) in prediction accuracy. This also corroborated the finding in previous studies, i.e., there might be no strict trade-off relationship between prediction fairness and accuracy.

2. Related work

2.1. Predictive tasks in education

With the advancement in technology development, various educational systems and tools have been developed and deployed to facilitate teaching and learning. The use of these systems and tools, on the other hand, has also enabled researchers to collect an unprecedented amount of fine-grained data describing every aspect of a student's learning process and performance (Bakhshinategh, Zaiane, ElAtia, & Ipperciel, 2018). For instance, according to Class Central,¹ platforms such as Coursera and edX have collected the learning trace data generated by over 220 million students during their learning with Massive Open Online Courses (MOOCs). With such a large amount of data collected, researchers have endeavored to make use of ML techniques to automate the processing of various educational tasks, including characterizing different types of posts generated by students in discussion forums (e.g., those relevant to course content vs. those not) (Ntourmas, Dimitriadis, Daskalaki, & Avouris, 2021; Wise, Cui, & Vytasek, 2016), modeling students' knowledge state and identifying potential knowledge gaps (Guzmán & Conejo, 2004; Mongkhonvanit, Kanopka, & Lang, 2019), evaluating students' written assignments (Hussein, Hassan, & Nassef, 2019; Ramesh & Sanampudi, 2021), identifying at-risk students (i.e., those in urgent need of help) (Dalipi, Imran, & Kastrati, 2018; Feng et al., 2019), predicting whether a student would pursue a career in STEM fields after college study (Almeda et al., 2020; Yeung & Yeung, 2019), and so on.

It is worth noting that these ML predictive models have been increasingly adopted as the backbone to facilitate educators to make decisions (Borrelli, Caballero-Caballero, & Ponce-Cueto, 2019, 2021; Herodotou, Maguire, McDowell, Hlosta, & Boroowa, 2021; Herodotou, Naydenova, Boroowa, Gilmour, & Rienties, 2020; Hlosta, Herodotou, Bayer, & Fernandez, 2021), e.g., what mark should be assigned to a student assignment or which students they should offer help first. For instance, Burgos et al. (2018) proposed to use a student's historical grade data as input to a logistic regression model to predict whether the student would drop out of a course or not. If yes, the instructors further took actions to provide support to the student (e.g., sending emails and making phone calls), through which the dropout rate was reduced by 14% compared to the previous academic year without any dropout prevention intervention. In a similar vein, Hlosta et al. (2021) constructed an ML-based predictive system to help instructors identify students that were at risk of failing, after which the instructors got in touch with the identified students to offer help. Through a randomized controlled study, this work demonstrated that, with the aid of this predictive system, the instructors were able to help more students to complete their studies. However, these studies generally ignored the potential prediction bias hidden behind those ML-based predictive systems (e.g., male students might be mistakenly predicted to be in need of more help than female students), and thus are limited in providing fair interventions to different groups of students.

2.2. Fairness-aware machine learning in education

As indicated before, prediction fairness is way more underrated than prediction accuracy in the application of ML techniques for educational predictive modeling. Representative studies in this strand of research can be found in Gardner, Brooks, and Baker (2019), Hutt, Gardner, Duckworth, and D'Mello (2019), Jiang and Pardos (2021), Kung and Yu (2020), Li, Xing, and Leite (2021), Tsai, Perrotta, and Gašević (2020), Yan et al. (2020), Yu, Lee, and Kizilcec (2021), which mostly focused on the measurement of prediction bias in the existing ML models adopted in education. For instance, Gardner et al. (2019) proposed a metric

¹ <https://www.classcentral.com/report/moocs-stats-and-trends-2021/>

called Absolute Between-ROC Area (ABROCA) to calculate the different prediction accuracy between groups of students as the bias displayed by an ML model. With ABROCA, Gardner et al. (2019) measured the prediction fairness of five common ML models used for predicting whether a student would be able to successfully complete a MOOC. In a broader view, Loukina et al. (2019) discussed several types of fairness that could be used to evaluate ML models used in education, based on which the authors further demonstrated the prediction bias displayed by the ML models used for scoring English language proficiency tests towards English-as-second-language students. In a different vein, Yu et al. (2021) investigated the benefit of including students' protective attributes as part of the input to empower the ML model for college dropout prediction and found that the use of students' protective attributes only marginally improved the prediction fairness of the ML models. To our knowledge, few studies attempted to develop methods to combat prediction bias by constructing fairness-aware ML for educational predictive tasks (Sha, Rakovic, Whitelock-Wainwright, et al., 2021). For example, Li et al. (2021) proposed that existing ML models can be orchestrated with the Seldonian algorithm (Thomas et al., 2019) (which could be regarded as a framework to guide ML models to behave desirably) to produce fairer prediction outcomes in predicting students' performance in math learning. In addition, Sha et al. (2022) demonstrated that, when classifying educational forum posts, the prediction bias could be attributed to the imbalance of the data used to train an ML model and class balancing techniques might be helpful in improving prediction fairness. Still, these studies were limited in that they mostly only focused on one specific predictive task in education, and thus it remained unknown whether the findings presented in these studies or the developed approaches were applicable in other educational tasks. Compared to these studies, our work distinguished itself by not only evaluating the prevalence of data bias (the main cause of prediction unfairness) but also proposing two simple but effective strategies to debias data for fairer predictions across a total of five different tasks to offer practical insights on building not only accurate but also fair ML models in education.

2.3. Techniques for debiasing data

In the broader research communities of ML and Data Mining, it has been well documented that the bias displayed by a predictive ML model can be partially attributed to the imbalanced nature of the data used to construct the model (Calmon, Wei, Vinzamuri, Natesan Ramamurthy, & Varshney, 2017; Friedler, Scheidegger, & Venkatasubramanian, 2014; Kamiran & Calders, 2012). By saying unbalanced, this often refers to the subtle patterns contained in the data, e.g., with a minority group being under-represented, which subsequently makes prediction errors produced on that group not only more likely but also less costly under certain accuracy measures. To combat data bias, several studies have been developed to *preprocess* the data before using it to train a model (Friedler et al., 2020; Jiang & Nachum, 2020; Li & Vasconcelos, 2019). For instance, Friedler et al. (2014) proposed to modify each protective attribute in the data so as to equalize the marginal distributions based on the subsets of the attribute with a given value. In addition to modify data attributes, an alternative is to apply class balancing techniques to equalize the distributions of data of different characteristics (Subramanian, Rahimi, Baldwin, Cohn, & Frermann, 2021; Yan et al., 2020; Yu, 2021). For instance, Yan et al. (2020) proposed a method called *fair class balancing* based on K-Means SMOTE, which makes no use of any information about protective attributes. By evaluating on two datasets, the proposed method was demonstrated to be capable of generating not only fair but also accurate prediction outcomes.

Driven by the success achieved in these studies, this study aimed to deliver a comprehensive evaluation of class balancing techniques in combating ML bias in educational predictive tasks and further explored how these techniques could be better utilized to deliver not only accurate but also fair predictions.

3. Methods

3.1. Tasks, datasets, and models

Here, we describe the tasks, the datasets, and the models we selected to replicate to tackle each task in this study. The task description and datasets statistics are summarized in Table 1. For each task, we selected a well-documented model which has been shown to achieve benchmark prediction accuracy performance in previous studies. By replicating these models, we expected to investigate to what extent their prediction accuracy could be maintained while being empowered to produce fairer predictions.

Forum Post Classification (Sha, Rakovic, Li, et al., 2021; Sha et al., 2023; Sha, Rakovic, Whitelock-Wainwright, et al., 2021). This dataset consisted of 3,703 posts authored by students in discussion forums of a Learning Management System Moodle at a university in Australia. Each post was labeled as *content-relevant* (e.g., "What is data literacy?") or *content-irrelevant* (e.g., "When is the due date of the first assignment?"). This dataset contained information not only about students' sexes but also their first-language backgrounds, which enabled us to investigate ML bias towards students from two different perspectives. In line with Sha, Rakovic, Li, et al. (2021), Sha et al. (2023), Sha, Rakovic, Whitelock-Wainwright, et al. (2021), we chose to couple the CNN-LSTM constructed in Guo et al. (2019), Wang, Yu, and Miao (2017) with the renowned pre-trained language model BERT (Devlin, Chang, Lee, & Toutanova, 2018) to build a classifier to distinguish the different types of posts in this dataset.

Legal Assignment Analysis. This dataset consisted of 5,800 assignments written by first-year students to analyze authentic legal cases at a university in Australia. When evaluating the quality of an assignment, instructors often paid particular attention to evaluate whether students successfully identify the Court Reasoning behind a legal case (i.e., sentences such as "*The appeal was argued on two grounds ...*" that were used to detail and analyze the reasoning of the Justice of the High Court). Each sentence in a student-written assignment was labeled as *relevant* or *irrelevant* to Court Reasoning. Similar to the forum dataset described above, this dataset also contained information both about students' sexes and first-language backgrounds. To build a classifier to characterize different categories of sentences, we used a variant of BERT called "*BERT Legal*", which was particularly fine-tuned on legal texts and shown to outperform the vanilla BERT in legal text classification (Chalkidis, Fergadiotis, Malakasiotis, Aletas, & Androutsopoulos, 2020). We coupled the *BERT Legal* model with a simple classification layer and fine-tuned the model as a whole on this dataset.

Student Dropout Prediction (Feng et al., 2019; Li et al., 2016). Originally, this dataset was released by XueTangX, a renowned MOOC platform in China, to fuel the KDD CUP competition organized in 2015.² This dataset consisted of students' fine-grained learning trace data in 39 courses on XueTangX, which detailed students' video-watching behaviors, their performance in assignments and tests, participation in discussion forums, and so on. With this dataset, the competition organizers expected to enable accurate ML models for predicting the likelihood of a student dropping out of a MOOC, which subsequently could be used by instructors to provide interventions to at-risk students and increase the retention rate. In addition to the learning trace data, this dataset also contained information about a student's sex. We replicated the Context-aware Feature Interaction Network model developed in Feng et al. (2019). In total, 22 features were engineered (e.g., the number of views in a lecture video and the number of correct responses made to quiz questions) to empower the model. These features were

² <https://kdd.org/kdd-cup>

Table 1

The description of the tasks included in this study and the dataset statistics. The numbers in the last four columns (i.e., Female, Male, English-second, English-first) specify the number of data instances that belong to students of a specific demographic attribute value.

Task	Description	Attributes	All	Demographic Distributions			
				Female	Male	English- second	English- first
Forum Post Classification	To classify to which category a discussion post belongs to, i.e., content-related vs. non-content-related	# Posts	3,703	2,225	1,478	2,110	1,593
		# Content-related	2,338	1,413	925	1,336	1,002
		Avg. words/post	131.39	138.90	116.77	119.51	146.00
		Avg. unique words/post	72.71	76.72	65.94	64.77	75.37
Legal Assignment Analysis	To identify sentences in student-written assignments that are used to detail the reasoning of the Justices of the High Court for authentic legal cases	# Sentences	23,347	14,803	8,544	7,945	15,402
		# Court reasoning sentences	9,167	5,799	3,368	3,082	6,072
		Avg. words/assignment	13.01	12.94	13.14	12.93	13.05
		Avg. unique words/assignment	11.94	11.89	12.03	11.82	12.00
Student Dropout Prediction	To predict whether a student would drop out of a MOOC	# Student enrollments	180,785	56,043	124,742	–	–
		# Dropout	148,118	42,954	105,164	–	–
		# Avg. log events/student	38.05	41.73	36.39	–	–
Student Performance Modeling	To predict the final performance of a student in a virtual learning environment	# Student enrollments	24,806	11,262	13,544	–	–
		# Students completed studies	12,362	5,743	6,619	–	–
		# Avg. log events/student	2.95	2.71	3.16	–	–
Career Prediction	To predict whether a student would pursue a career in STEM fields	# Students	591.00	247.00	344.00	–	–
		# Students pursued a STEM career	125.00	47.00	78.00	–	–
		# Avg. log events/student	536.34	537.18	535.73	–	–

augmented, embedded, and fused by using a context-smoothing structure and then served as input to a CNN model to produce a prediction outcome.

Student Performance Modeling (Kuzilek, Hlosta, & Zdrahal, 2017). This dataset consisted of the learning trace data generated by 24,806 students in 22 courses in a virtual learning enrollment, which was developed by the Open University in the UK. The information about a student's sex was also included in the dataset. The release of this dataset was endowed with the mission to spur the construction of ML models which could be used to accurately predict students' final performance in a course, i.e., whether a student would *pass* or *fail* the course. In line with the OULAD project,³ i.e., a project particularly created to analyze this dataset, we engineered a total of 112 features from student data (e.g., the number of attempts made in answering a quiz question) as input to a Random Forest model for prediction.

Career prediction (Liu et al., 2020; Yeung & Yeung, 2019). This dataset was released to enable the prediction of whether a student would pursue a STEM career after college study, which consisted of not only 591 middle-school students' learning traces in ASSISTments (a well-known educational platform for math learning in Northern America) but also information about their sexes and affective states based on students' learning history on the platform. In line with Yeung and Yeung (2019), we adopted an extended version of the Deep Knowledge Tracing model proposed in Yeung and Yeung (2018) to model a student's knowledge state, which was then used together with the student's profile information as input to a logistic regression model to generate a prediction outcome.

For simplicity, we denote these tasks as Forum, Legal, Dropout, Performance, and Career, respectively, in the remainder of this article.

3.2. Data bias measures

Recall that part of our goal was to diagnose bias hidden behind datasets used for constructing predictive ML models in education. As suggested by Smith, Martinez, and Giraud-Carrier (2014), Yan et al. (2020), we adopted the following metrics to measure data bias from two different perspectives:

Distribution bias (*Dist-bias* for short), which measures distribution difference of a specific task label (e.g., *pass* or *fail* in modeling student performance). Formally, this metric is defined as the difference of probabilities of prediction task label $Y = y$, conditioned upon the value of a protective attribute G where $G \in \{0, 1\}$:

$$D(y) = Pr(Y = y | G = 1) - Pr(Y = y | G = 0), \quad (1)$$

where $Y \in \{0, 1\}$. To account for the distribution biases calculated based on different values of the task label $Y \in \{0, 1\}$, we averaged $D(Y = 0)$ and $D(Y = 1)$ as the overall distribution bias of a dataset in later analyses. The smaller the distribution bias a dataset has, the more likely a predictive model built on the dataset would produce fair predictions.

Hardness bias (*Hard-bias* for short), which measures the degree to which data instances contained in a dataset are difficult to be correctly labeled. The core idea of this metric is that, if a data instance do not share the same task label with most of its *k-nearest neighbors*, then it tends to be difficult to correctly label this instance. Similar to Smith et al. (2014), Yan et al. (2020), we used the metric *k-Disagreeing Neighbors* (kDN) to measure the local overlap of a data instance x with its k -nearest neighbors (identified by calculating their Euclidean distance) regarding their task labels. A large kDN (close to 1) indicates that the data instance is difficult to be correctly classified. We chose $k = 5$ to calculate the kDN of an instance (as suggested in Yan et al. (2020)). If the kDN distribution of a student group (e.g., female) is larger than that of the other group (e.g., male), there exists Hard-bias (denoted as H) between the two groups, which can be calculated by

³ https://github.com/gogoladzetado/Open_University_Analytics

applying the Jensen–Shannon (denoted as J) distance (a symmetric version of Kullback–Leibler divergence with a finite value), as defined below:

$$H(y) = J(f(\{kDN(x, y) \mid G = 1\}) - f(\{kDN(x, y) \mid G = 0\})) \quad (2)$$

where $G \in \{0, 1\}$, $Y \in \{0, 1\}$, and $f(\{kDN(x, y) \mid G = g\})$ is the kDN density distribution of data instances with $G = g$ and $Y = y$. Similar to Dist-bias, we averaged $H(Y = 0)$ and $H(Y = 1)$ as the overall Hard-bias of a dataset.

3.3. Methods for debiasing data

Here, we describe the methods we proposed to extend class balancing techniques to enable ML models to generate fairness-aware prediction outcomes towards students regardless of their different protective attribute values, as depicted in Fig. 1. As investigated in prior studies (Sha, Rakovic, Whitelock-Wainwright, et al., 2021; Subramanian et al., 2021; Yan et al., 2020; Yu, 2021), class balancing techniques (including both over-sampling and under-sampling) mainly balance the uneven distribution of data instances of different tasks labels (① and ② in Fig. 1). Though being demonstrated to be useful in certain cases, the pure application of such class balancing techniques cannot guarantee the reduction of the distribution bias of a dataset (as the balancing was performed upon one single dimension, i.e., the task label) and thus often lead to a greater prediction bias (Yan et al., 2020). Therefore, we proposed that the demographic labels of students should also be taken into account (③ in Fig. 1). That is, the data specific to different protective groups $G \in \{0, 1\}$ should be equal not only within a task label but also across different task labels $Y \in \{0, 1\}$. When applying over-sampling techniques, the group ($G = g$ and $Y = y$) with the most data instances is used as the reference group, i.e., other groups should be over-sampled to have the same number of data instances; while under-sampling techniques are used, the group with the least data instances is used as the reference group. Though the use of both the task and demographic labels can eliminate distribution bias in a dataset, the re-sampled dataset might have a higher Hard-bias than the original dataset, which poses threats to not only an ML model's prediction fairness across different student groups but also its prediction accuracy within a specific student group. Thus, we further introduced a simple (but effective, as demonstrated later) sample selection step (④ in Fig. 1), i.e., we repeated the balancing process with the use of both the task and demographic labels K times, after which we calculate the Hard-bias of the resulted K re-sampled datasets and selected the one with the lowest Hard-bias to train an ML model. It should be noted that, in practice, due to the different nature of different datasets (some are very imbalanced while the others are relatively more balanced), the overall Hard-bias of the resulted re-sampled dataset is not necessarily lower than that of the original one. An alternative approach is to incorporate a *lower Hard-bias constraint* into the balancing process in an instance-by-instance manner. For instance, when applying over-sampling techniques, we may determine whether to include a newly-generated instance to the dataset by measuring whether the inclusion of this instance would increase or decrease the overall Hard-bias of the dataset. However, this alternative approach can be costly in computation when dealing with large-scale datasets. Therefore, we first try the simple sample selection (step ④) by repeating the balancing process with K times. If a re-sampled dataset with a lower Hard-bias cannot be attained, then we apply the instance-by-instance approach to ensure a lower Hard-bias.

To summarize, we implemented and compared the following methods to use class balancing techniques for constructing fairness-aware ML models:

- **Method-1**, i.e., the pure application of a class balancing technique, which only balances the data distribution according to the task label (i.e., ① and ② in Fig. 1);

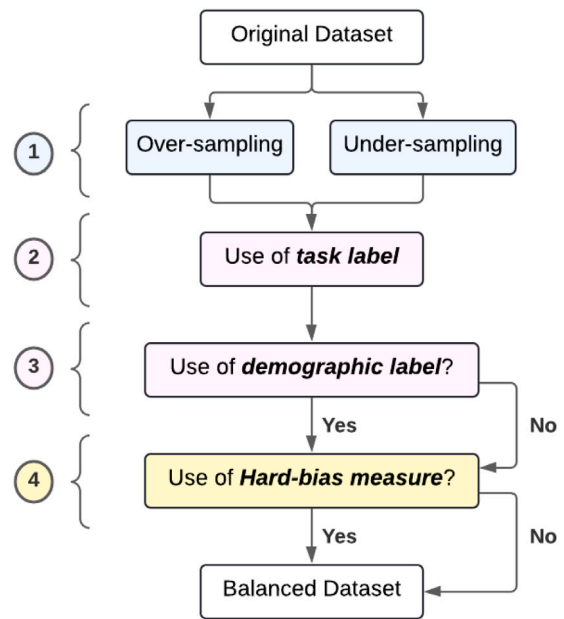


Fig. 1. Workflow of the proposed data balancing method. The use of ② and ③ is to tackle Distribution bias, while ④ is to tackle Hardness bias.

- **Method-2**, which balances a dataset by taking both the task and demographic labels into account (i.e., ①, ②, and ③ in Fig. 1);
- **Method-3**, which not only uses the task and demographic labels to balance a dataset but also adopts an additional sample selection step to lower the overall Hard-bias of the dataset (i.e., ①, ②, ③, and ④ in Fig. 1).

For the selection of class balancing techniques, we selected two representative techniques, i.e., SMOTE (Chawla, Bowyer, Hall, & Kegelmeyer, 2002) for over-sampling and NearMiss (Mani & Zhang, 2003) for under-sampling, as the testbeds to evaluate the effectiveness of Method-1/2/3 described above.

3.4. Study setup

Data pre-processing. For each task, we randomly split the data in the ratio of 8:2 as the training and testing set for model construction. During data balancing, the training data samples are added or removed based on different balancing approaches (Method-1, 2, and 3) as detailed in Section 3.3. For Method-1, we balanced the number of samples to achieve an equal sample distribution of different task labels in the training set. So when over-sampling CBTs were used, we added samples to the minority class until the sample number reached parity with the majority class. Take the Forum dataset for example, the non-content-related posts were over-sampled to the number of samples of the content-related posts. For Method-2 and Method-3, the number of samples were balanced according to demographic and task labels. For instance, when over-sampling the Forum (S) dataset, we applied CBTs to over-sample male content-related, male non-content-related and female non-content-related posts until the number of samples reached parity with the number of samples in female content-related posts. Similarly, in under-sampling, we removed samples of female content-related, female non-content-related, and male content-related to reach parity with the number of male non-content-related posts. Importantly, the test set remained unchanged during the later over- and under-sampling experiments. When training a model, we performed parameter tuning by 5-fold cross-validation and applied the parameter value that gave the lowest cross-validation average error. To pre-process the textual data used in the tasks of Forum Post Classification

Table 2

Results on dataset bias and the prediction performance of equipping SMOTE with the proposed data balancing methods. Demographic labels of sex (S) and first-language background (L) were used to balance the data, as indicated in the Dataset column. Original column indicates the use of the original dataset without applying any data balancing methods; while the Balanced-1/2/3 columns refer to the re-sampled datasets after using the Method 1/2/3 described in Section 3.3, respectively. The numbers within brackets indicate the increased/decreased performance compared to Original. The best result within each row is marked with ‡.

Dimension	Metric	Row ID	Dataset	Original	Fair-CBT	Balanced-1	Balanced-2	Balanced-3
Data Bias	Dist-bias	1	Forum (S)	0.1001	0.0647 (35.36%)	0.1113 (−11.19%)	‡ 0 (100%)	‡ 0 (100%)
		2	Dropout (S)	0.1920	0.2670 (−39.09%)	0.2024 (−5.42%)	‡ 0 (100%)	‡ 0 (100%)
		3	Performance (S)	0.0447	0.0329 (26.49%)	0.0448 (−0.22%)	‡ 0 (100%)	‡ 0 (100%)
		4	Legal (S)	0.1403	0.1365 (2.70%)	0.1610 (−14.75%)	‡ 0 (100%)	‡ 0 (100%)
		5	Career (S)	0.0826	0.0594 (28.08%)	0.1288 (−55.93%)	‡ 0 (100%)	‡ 0 (100%)
		6	Forum (L)	0.0738	0.1105 (−49.69%)	0.0821 (−11.25%)	‡ 0 (100%)	‡ 0 (100%)
		7	Legal (L)	0.1612	0.1335 (17.18%)	0.1693 (−5.02%)	‡ 0 (100%)	‡ 0 (100%)
	Hard-bias	8	Forum (S)	0.2968	0.3162 (−6.54%)	0.3099 (−4.41%)	0.2772 (6.60%)	‡ 0.2632 (11.32%)
		9	Dropout (S)	0.3530	0.3499 (0.87%)	0.3618 (−2.49%)	0.3133 (11.25%)	‡ 0.3098 (12.24%)
		10	Performance (S)	0.5671	0.5622 (0.86%)	0.5614 (1.01%)	0.5912 (−4.25%)	‡ 0.5601 (1.23%)
		11	Legal (S)	0.3662	0.3573 (2.43%)	0.3242 (11.47%)	0.3532 (3.55%)	‡ 0.3302 (9.83%)
		12	Career (S)	0.4787	0.4539 (5.18%)	0.5927 (−23.81%)	0.6223 (−30.00%)	‡ 0.4524 (5.49%)
		13	Forum (L)	0.3062	0.3332 (−8.82%)	0.3101 (−1.27%)	0.2939 (4.02%)	‡ 0.2907 (5.06%)
		14	Legal (L)	0.3533	0.3120 (11.68%)	0.3541 (−0.23%)	0.3201 (9.40%)	‡ 0.3109 (12.00%)
PredictionPerformance	ABROCA	15	Forum (S)	0.0406	0.0298 (26.50%)	0.0455 (−12.07%)	0.0264 (34.98%)	‡ 0.0135 (66.75%)
		16	Dropout (S)	0.0109	0.0107 (1.97%)	0.0144 (−32.11%)	0.0105 (3.67%)	‡ 0.0066 (39.45%)
		17	Performance (S)	0.0141	0.0154 (−9.21%)	0.0145 (−2.84%)	‡ 0.0139 (1.42%)	0.0140 (0.71%)
		18	Legal (S)	0.0124	0.0138 (−11.56%)	0.0133 (−7.26%)	0.0087 (29.84%)	‡ 0.0066 (46.77%)
		19	Career (S)	0.0858	0.0753 (12.22%)	0.1079 (−25.76%)	0.0678 (20.98%)	‡ 0.0637 (25.76%)
		20	Forum (L)	‡ 0.0195	0.0204 (−4.50%)	0.0265 (−35.90%)	0.0233 (−19.49%)	0.0205 (−5.13%)
		21	Legal (L)	0.0130	0.0119 (8.39%)	0.0084 (35.38%)	0.0127 (2.31%)	‡ 0.0066 (49.23%)
	AUC	22	Forum (S)	0.9462	0.9584 (1.28%)	0.9485 (0.24%)	0.9402 (−0.63%)	‡ 0.9607 (1.53%)
		23	Dropout (S)	0.8878	‡ 0.8928 (0.56%)	0.8884 (0.07%)	0.8892 (0.16%)	0.8896 (0.20%)
		24	Performance (S)	0.8852	0.8648 (−2.30%)	0.8838 (−0.16%)	‡ 0.8863 (0.12%)	0.8862 (0.11%)
		25	Legal (S)	0.8815	0.8817 (0.02%)	‡ 0.8846 (0.35%)	0.8807 (−0.09%)	0.8801 (−0.16%)
		26	Career (S)	0.6652	0.6523 (−1.93%)	0.6434 (−3.28%)	0.6929 (4.16%)	‡ 0.6970 (4.78%)
		27	Forum (L)	‡ 0.9639	0.9625 (−0.14%)	0.9633 (−0.06%)	0.9627 (−0.12%)	0.9619 (−0.21%)
		28	Legal (L)	‡ 0.8875	0.8702 (−1.95%)	0.8834 (−0.46%)	0.8763 (−1.26%)	0.8796 (−0.89%)

and Legal Assignment Analysis, we removed all invalid characters and stopwords, and then applied word-stemming using the Python package *NLTK*.⁴

Method comparison. To demonstrate the effectiveness of the class balancing approach developed in this study, we selected the benchmarking method Fair Class Balancing Technique (denoted as Fair-CBT) developed in Yan et al. (2020) as a baseline for comparison. When implementing Fair-CBT, we followed the same steps as suggested in Yan et al. (2020): (i) K-means clustering was first implemented in the original dataset to obtain sample clusters based on the average silhouette score; (ii) samples close to the cluster boundaries were removed; (iii) when an over-sampling technique was used (i.e., SMOTE), new minority class samples were generated within each cluster until samples sizes of different classes reached parity; when an under-sampling technique was used (i.e., NearMiss), the majority class samples were removed within each cluster until samples sizes of different classes were equal.

Model Implementation and Training. The implementation of the ML models is detailed below. (i) For *Forum Post Classification*, the CNN-LSTM model was implemented using *tensorflow* and the BERT embeddings of input text were generated using *Bert-as-service*.⁵ The model parameters were set in accordance with previous studies (Clavié & Gal, 2019; Guo et al., 2019; Wei, Lin, Yang, & Yu, 2017). (ii) For

Legal Assignment Analysis, we adopted the open-sourced *Legal BERT* model (Chalkidis et al., 2020), whose parameters were further fine-tuned by following the procedure developed in Chalkidis et al. (2020). (iii) For *Student Dropout Prediction*, we replicated the model by using its open-sourced GitHub repository.⁶ (iv) For *Student Performance Modeling*, we first replicated the model by using its open-sourced GitHub repository⁷ and then fine-tuned its parameters by using grid search. (v) For *Career Prediction*, we replicated the model by using its open-sourced GitHub repository⁸ and followed the parameter setting specified in the repository. When implementing the Method-3 described in Section 3.3, we set $K = 100$. For further details about the implementation and training process, please refer to our open-sourced GitHub repository via <https://github.com/lsha49/FairEdu>, which contains all data and code used in this study for replication.

Evaluation metrics To measure data bias, we calculated the Dist-bias and Hard-bias measures (described in Section 3.2). To measure an ML model's prediction accuracy, we adopted the widely-used metric Area Under the Curve (AUC) (Dalipi et al., 2018; Guo et al., 2019; Li et al., 2021; Yeung & Yeung, 2019); for prediction fairness, we used the metric ABROCA (Gardner et al., 2019), which is a group-level fairness measure based on equalized odds and calculated by measuring the absolute difference between the ROC curves of two student groups

⁴ <https://www.nltk.org/>

⁵ <https://github.com/hanxiao/bert-as-service>

⁶ https://github.com/wzfaha/dropout_prediction

⁷ https://github.com/gogoladzetedo/Open_University_Analytics

⁸ <https://github.com/ckyeungac/ADM2017>

Table 3

Results on dataset bias and the prediction performance of equipping NearMiss. The numbers within brackets indicate the increased/decreased performance compared to Original. The best result within each row is marked with ‡.

Dimension	Metric	Row ID	Dataset	Original	Fair-CBT	Balanced-1	Balanced-2	Balanced-3
Data Bias	Dist-bias	1	Forum (S)	0.1001	0.0718 (28.23%)	0.1090 (−8.89%)	‡ 0 (100%)	‡ 0 (100%)
		2	Dropout (S)	0.1920	0.1758 (8.46%)	0.2010 (−4.69%)	‡ 0 (100%)	‡ 0 (100%)
		3	Performance (S)	0.0447	0.0030 (93.40%)	0.0447 (0.00%)	‡ 0 (100%)	‡ 0 (100%)
		4	Legal (S)	0.1403	0.1418 (−1.04%)	0.1437 (−2.42%)	‡ 0 (100%)	‡ 0 (100%)
		5	Career (S)	0.0826	0.1583 (−91.69%)	0.1288 (−48.67%)	‡ 0 (100%)	‡ 0 (100%)
		6	Forum (L)	0.0738	0.0864 (−17.02%)	0.0720 (2.44%)	‡ 0 (100%)	‡ 0 (100%)
		7	Legal (L)	0.1612	0.1342 (16.74%)	0.1680 (−4.22%)	‡ 0 (100%)	‡ 0 (100%)
	Hard-bias	8	Forum (S)	0.2968	0.2930 (1.28%)	0.3065 (−3.27%)	0.2880 (2.96%)	‡ 0.2839 (4.35%)
		9	Dropout (S)	0.3530	0.3418 (3.19%)	0.3311 (6.20%)	0.3083 (12.66%)	‡ 0.2933 (16.91%)
		10	Performance (S)	0.5671	0.5659 (0.21%)	0.5654 (0.30%)	0.5665 (0.11%)	‡ 0.5642 (0.51%)
		11	Legal (S)	0.3662	0.3142 (14.21%)	0.3155 (13.84%)	0.3157 (13.79%)	‡ 0.3019 (17.56%)
		12	Career (S)	0.4787	0.4289 (10.41%)	0.6514 (−36.08%)	0.4399 (8.11%)	‡ 0.3607 (24.65%)
		13	Forum (L)	0.3062	0.2421 (20.93%)	0.3044 (0.59%)	0.2844 (7.12%)	‡ 0.2321 (24.20%)
		14	Legal (L)	0.3533	0.3619 (−2.43%)	0.3555 (−0.62%)	0.3512 (0.59%)	‡ 0.3223 (8.77%)
PredictionPerformance	ABROCA	15	Forum (S)	0.0406	0.0500 (−23.17%)	0.0413 (−1.72%)	0.0260 (35.96%)	‡ 0.0213 (47.54%)
		16	Dropout (S)	0.0109	0.0102 (6.35%)	0.0121 (−11.01%)	0.0092 (15.60%)	‡ 0.0088 (19.27%)
		17	Performance (S)	0.0141	0.0156 (−10.85%)	0.0142 (−0.71%)	‡ 0.0135 (4.26%)	0.0138 (2.13%)
		18	Legal (S)	0.0124	0.0086 (30.63%)	0.0142 (−14.52%)	0.0075 (39.52%)	‡ 0.0064 (48.39%)
		19	Career (S)	0.0858	‡ 0.0743 (13.40%)	0.1123 (−30.89%)	0.1202 (−40.09%)	0.0832 (3.03%)
		20	Forum (L)	0.0195	0.0180 (7.48%)	0.0202 (−3.59%)	0.0224 (−14.87%)	‡ 0.0176 (9.74%)
		21	Legal (L)	0.0130	0.0111 (14.24%)	0.0124 (4.62%)	0.0051 (60.77%)	‡ 0.0046 (64.62%)
	AUC	22	Forum (S)	0.9462	0.9461 (−0.01%)	0.9422 (−0.42%)	‡ 0.9501 (0.41%)	0.9411 (−0.54%)
		23	Dropout (S)	0.8878	0.8379 (−5.62%)	0.8879 (0.02%)	0.8902 (0.27%)	‡ 0.8922 (0.50%)
		24	Performance (S)	‡ 0.8852	0.8649 (−2.30%)	0.8808 (−0.50%)	0.8769 (−0.94%)	0.8819 (−0.37%)
		25	Legal (S)	0.8815	0.8783 (−0.36%)	0.8824 (0.10%)	0.9283 (5.31%)	‡ 0.9296 (5.46%)
		26	Career (S)	‡ 0.6652	0.6319 (−5.00%)	0.6131 (−7.83%)	0.5983 (−10.06%)	0.6256 (−5.95%)
		27	Forum (L)	‡ 0.9639	0.9617 (−0.23%)	0.9625 (−0.15%)	0.9473 (−1.72%)	0.9413 (−2.34%)
		28	Legal (L)	‡ 0.8875	0.8824 (−0.58%)	0.8828 (−0.53%)	0.8870 (−0.06%)	0.8807 (−0.77%)

(e.g., female and male) at all thresholds. ABROCA has been used in several prior studies to measure the prediction fairness of ML models used in education.

4. Results

The results of using SMOTE as a testbed are given in Table 2, and those of NearMiss can be found in Table 3. These results can be discussed from several aspects, as detailed below.

The prevalence of data bias. In line with our prior study (Sha, Rakovic, Whitelock-Wainwright, et al., 2021), we found data bias generally manifested itself in terms of both Dist-bias and Hard-bias in all educational datasets (i.e., the datasets which were frequently adopted in longstanding educational predictive tasks) used in this study (Row 1–14, Column *Original* in Table 2). The Dist-bias values ranged from [0.0447, 0.1920], while the Hard-bias values ranged from [0.2968, 0.5671]. In particular, though Performance(S) having the lowest Dist-bias (0.0447), its Hard-bias was the largest in all datasets (up to 0.5671). This suggests that the bias hidden behind a dataset should be measured from more than one perspective.

The power of vanilla class balancing techniques. By scrutinizing the *Balanced-1* results in Tables 2 and 3, we can conclude that, by simply balancing data distribution upon the dimension of prediction task label, there was no guarantee that a less-biased dataset could be achieved. In fact, when applying the over-sampling technique (SMOTE), the Dist-bias was increased in all tasks (Row 1–7, Table 2) and five out of the seven evaluation tasks also experienced an increase in Hard-bias. In particular, the Dist-bias and Hard-bias of Career(S) worsened

by 55.93% and 23.81%, respectively. Similarly, when applying under-sampling (NearMiss) to balance dataset, five out of seven tasks had worsened Dist-bias and three out of seven tasks had worsened Hard-bias. This corroborated with the findings presented in Yan et al. (2020), i.e., the vanilla class balancing technique tended to exacerbate the existing bias in datasets. Correspondingly, most of the ML models built on these re-sampled datasets did not gain improvements in prediction fairness. To our surprise, the accuracy performance of these models even slightly decreased up to 3.28% in four out of the seven evaluation tasks when applying over-sampling (measured by AUC in Row 22–28 in Table 2), and up to 7.83% in five out of seven tasks when applying under-sampling (Table 3).

The importance of demographic label. In contrast, by incorporating demographic labels together with task labels to eliminate the dist-bias of a dataset (i.e., Method-2 described in Section 3.3), we can observe that five out of the seven evaluation tasks had 3.55% ~ 11.25% improvement in Hard-bias (Row 8–14, Column *Balanced-2* in Table 2). More importantly, six out of seven ML models trained on these *Balanced-2* datasets had an improvement (1.42% ~ 34.98%) in prediction fairness measured by ABROCA (Row 15–21, Column *Balanced-2* in Table 2). At the same time, only four of these ML models had a smaller decrease of 0.09% ~ 1.26% in prediction accuracy (Row 22–28, Column *Balanced-2* in Table 2), compared to the results derived on *Balanced-1*. We observed similar patterns in under-sampling results Table 3, i.e., data bias and prediction bias were improved in *Balanced-2* column. This provided strong evidence on the necessity of incorporating students'

demographic labels to eliminate Dist-bias in datasets before using them as input to train ML models.

The impact of Hard-bias. Similar to *Balanced-2*, *Balanced-3* also eliminated Dist-bias by using both task and demographic labels to balance a dataset. In addition, a sample selection step was introduced (i.e., the Method-3 described in Section 3.3) to lower the overall Hard-bias of the re-sampled dataset. In the following, let us first delve into the over-sampling results presented in Table 2. It should be noted that all of the seven evaluation tasks achieved their best Hard-bias results compared to other methods (Row 8–14, Column *Balanced-3* in Table 2). As a result, most of the evaluation tasks (six out of seven) achieved better prediction fairness and, particularly, five displayed the fairest prediction results, i.e., Forum (S), Dropout (S), Legal (S), Career (S), and Legal (L) (Row 15–21, Column *Balanced-3* in Table 2). In particular, the fairness improvement displayed in Forum(S), Legal (S) and Legal (L) was around 50%. The only evaluation task with no fairness improvement is Forum (L) (Row 20, Column *Balanced-3* in Table 2). This implied that, in addition to the Dist-bias and Hard-bias hidden behind the training data, there may be other factors contributing to the predictive unfairness, e.g., bias occurred during the annotation or training process. We leave the exploration of such factors in our future work. When delving into the results on prediction accuracy (Row 22–28, Column *Balanced-3* in Table 2), we noticed that four out of the seven evaluation tasks achieved better AUC accuracy as well; while the remaining three had a minimal increase/decrease of less than 1%.

On the other hand, similar findings can be found in the under-sampling results presented in Table 3. For instance, when scrutinizing the ABROCA results (Row 15–21, Column *Balanced-3* in Table 3), we observe that all evaluation tasks achieved improvement of prediction fairness compared to the *Original* results and, in particular, five of them achieved the fairest prediction results, with an improvement up to 64%. Similarly, most of the evaluation tasks only had a relatively small decrease in prediction accuracy.

These results together demonstrated that the imbalance of a dataset could be further alleviated by using the sample selection step we introduced, with which an ML model could be empowered to generate much fairer prediction outcomes at the expense of a minimal drop in prediction accuracy. Overall, we did not observe an explicit trade-off relationship between prediction fairness and accuracy, i.e., the improvement of prediction fairness does not necessarily come along with a performance drop in prediction accuracy.

Superiority over Fair-CBT. As shown in both Tables 2 and 3 (Row 15–21, Column *Fair-CBT*), the benchmarking method Fair-CBT did not necessarily achieve fairer prediction results. Specifically, compared to the *Original* results, Fair-CBT were fairer in only four out of seven evaluation tasks in the over-sampling results (Table 2), while it is only five in the under-sampling results (Table 3). When being compared to *Balanced-3*, Fair-CBT was inferior to *Balanced-3* in 12 out of the 14 over- and under-sampling evaluation settings, with only exceptions in the over-sampling Forum (L) and under-sampling Career (S). This further demonstrated the robustness of *Balanced-3* in debiasing data for not only accurate but also fair predictions.

5. Discussion and conclusion

As ML models have been increasingly used to tackle various educational predictive tasks (Lee, 2018; Li et al., 2016; Sha et al., 2023), reaching a high predictive accuracy alone is often insufficient as educational researchers endeavor to ensure equal learning opportunities across different student population subgroups, especially those that are traditionally disadvantaged (e.g., female students in higher education (Lin, 2016)). Driven by this, we systematically investigated data bias from two different perspectives (i.e., distribution and hardness) to

which predictive unfairness may be potentially attributed. We quantified the distribution and hardness bias of a total of five popular datasets used for predictive tasks in the educational domain, and proposed CBT-based sampling methods to improve the prediction fairness of ML models. To our knowledge, our study is the first to extend the use of CBTs to alleviate the impact of biased datasets for predictive modeling in education and deliver the largest scale measurement of data bias on educational datasets. Here, we further elaborated on our study's practical insights and implications and discussed the limitations to be addressed in the future.

5.1. Implications

Firstly, we observed that all educational datasets used in this study exerted bias to a certain extent. For instance, in the datasets used for the tasks of *Student Dropout Prediction* and *Legal Assignment Analysis*, there existed a significant difference regarding the distribution of data instances specific to students of different protective attributes. More importantly, a dataset can be characterized as being relatively balanced in one data bias measure but being extremely imbalanced in another measure (e.g., the dataset for *Student Performance Modeling*). We demonstrated that the prediction bias displayed by ML models built on these datasets could be largely attributed to the imbalanced nature of these datasets. Therefore, researchers and practitioners should scrutinize potential biases hidden in datasets from different perspectives and apply appropriate methods to alleviate the impact of these data biases before using the datasets to train ML models. In addition, given the increasing importance of open-sourced datasets for educational researchers, the curators and publishers of these datasets should perform similar practices to minimize the impact of biased datasets on relevant research works.

Secondly, our evaluation results on *Balanced-1* showed that the vanilla class balancing techniques (i.e., balancing data samples by the task label) tended to exacerbate existing data and predictive bias. Instead, by using both task and demographic labels (*Balanced-2*) to eliminate Dist-bias in a dataset, the prediction fairness generally gained some improvement (in five out of seven datasets with an ABROCA improvement up-to 61%). This may be due to the fact that data imbalance among different demographic groups caused model to be under-trained on certain minority subgroup (e.g., female students) compared to the majority group (male students). This finding highlighted that Dist-bias could be an important cause of predictive bias.

Thirdly, after removing Dist-bias, it is possible to further improve fairness by simply selecting a re-sampled dataset with a lower Hard-bias than the original dataset. We demonstrated the benefits of doing so, i.e., the *Balance-3* method, by comparing it with a benchmarking fairness-oriented class balancing method called Fair-CBT, and showed that our method achieved not only the best prediction fairness (with an ABROCA improvement of up-to 66% compared to the baseline) but also a comparable prediction accuracy in most of the settings. However, the effectiveness of such a sample selection strategy might depend on the nature of the dataset and should be further evaluated. Specifically, the level of demographic disparity (in terms of distribution and hardness dimensions) could differ among different datasets, which may lead to different numbers of samples being generated/removed. For instance, for the task of *Student Performance Modeling*, only 11% of the re-sampled dataset was generated to reach parity in the distribution dimension, which did not bring a significant improvement in the hardness dimension. Consequently, the improvement on prediction fairness was limited.

5.2. Limitations

Firstly, our fairness evaluations of ML models were based on two types of protective attributes (i.e., sex and first-language background) in this study. It is worth further investigating data bias by including

other types of protective attributes (e.g., age and highest education level) and, more importantly, measuring bias by taking more than one protective attribute into account at a time. Secondly, we only included two class balancing techniques (i.e., SMOTE and NearMiss) in this study. Although these two strategies are representatives of the respective categories of over-sampling and under-sampling techniques, we acknowledge that there are other types of class balancing techniques proposed for problems in other scenarios, e.g., tackling multi-class classification problems (Sahare & Gupta, 2012) and working with severely-unbalanced datasets (García, Sánchez, & Mollineda, 2012). In the future, it would be helpful to combine the data balancing methods developed in this study with other types of class balancing techniques to tackle more complex educational predictive tasks. Lastly, the effectiveness of the data balancing methods developed in this study necessitates the explicit use of students' protective attributes. It remains still disputable whether such protective attributes should be used in building ML models. Therefore, it is worth designing and evaluating data balancing methods that do not rely on the explicit use of students' protective attributes.

CRedit authorship contribution statement

Lele Sha: Conceptualization, Experimentation, Results analysis, Research implications, Manuscript writing. **Dragan Gašević:** Supervision, Manuscript writing. **Guanliang Chen:** Supervision, Conceptualization, Research implications, Manuscript writing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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